PREDICTIVE ANALYTICS: MACHINE LEARNING PROJECT REPORT

(Project Semester August-December 2026)

Online Shoppers Purchasing Intention Prediction Using Machine Learning

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Programme and Section: B.Tech CSE(P132), K2P24NK

Course Code: INT-234

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CERTIFICATE

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DECLARATION

I, Aditya Kumar, student of B.Tech, under CSE/IT Discipline at, Lovely Professional University, Punjab, hereby declare that all the information furnished in this project report is based on my own intensive work and is genuine.

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ACKNOWLEDGEMENT

I express my profound gratitude and sincere appreciation to my esteemed project supervisor, Dr. Mrinalini Rana (UID: 22138), Associate Professor, School of Computer Science and Engineering, Lovely Professional University, for her constant guidance, academic mentorship, and insightful feedback throughout the duration of this research project. Her profound knowledge in Data Science and Machine Learning has been a continuous source of intellectual stimulation and methodological discipline.

I extend my heartfelt thanks to the Lovely School of Computer Science and Engineering and Lovely Professional University, Phagwara, for offering an exceptional academic environment and providing access to the necessary high-performance computational resources, data analysis software, and institutional digital libraries that made the execution of this machine learning research possible.

I also take this opportunity to convey my appreciation to the faculty members and technical coordinators of the Discipline of CSE/IT for their constructive advice, technical suggestions, and institutional cooperation. Special thanks are due to my peers and fellow researchers for their engaging discussions, critical reviews, and cooperation during various analytical and empirical milestones of this project.

Finally, I owe an immense debt of gratitude to my family and loved ones for their enduring patience, moral encouragement, and unconditional belief in my capabilities throughout this project semester.

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Table of Contents

Section No.	Title	Page No.
1	**Introduction**	7
1.1	Project Overview and E-Commerce Landscape	7
1.2	Problem Statement: The Conversion Chasm in Online Retail	7
1.3	Research Objectives	8
1.4	Significance and Business Value of Study	9
1.5	Scope and Delimitations of the Project	9
1.6	Architectural Workflow and Methodology	10
2	**Source of Dataset**	11
2.1	Dataset Provenance and Academic Attribution	11
2.2	Feature Dictionary and Attribute Schema	11
2.3	Feature Categorization: Behavioral, Demographic, Temporal	12
2.4	Summary Statistics and Empirical Distributions	13
2.5	Target Variable Characteristics and Class Imbalance	13
3	**EDA Process and Data Preprocessing Pipeline**	15
3.1	Initial Ingestion and Exploratory Structural Audit	15
3.2	Duplicate Record Detection and Data Cleaning	15
3.3	Domain Constraint and Anomaly Validation	16
3.4	Feature Engineering and Categorical One-Hot Encoding	16
3.5	Stratified Train-Test Splitting Methodology	16
3.6	Dual Feature Scaling Architectures (Standard vs Min-Max)	17

4	**Analysis on Dataset**	18
4.1	Overall Session Purchase Conversion Analysis	18
4.2	Visitor Type Behavioral Loyalty and Conversion Dynamics	21
4.3	Page Category Browsing Duration and User Engagement	23
4.4	Temporal Seasonality and Monthly Purchase Trends	25
4.5	Day-of-Week and Weekend Purchasing Propensity	27
4.6	Event Proximity and Special Day Purchasing Dynamics	29
4.7	Predictive Classification via Regularized Logistic Regression	31
4.8	Probabilistic Modeling via Gaussian Naive Bayes Classifier	34
4.9	Instance-Based Classification via K-Nearest Neighbors (KNN)	36
4.10	Non-Parametric Rule Extraction via Decision Tree Classification	38
4.11	Multi-Model Benchmark and Algorithmic Performance Evaluation	40
5	**Conclusion**	43
5.1	Synthesis of Empirical Findings	43
5.2	Strategic Recommendations for E-Commerce Platforms	44
5.3	Comparative Model Selection Guidelines	44
6	**Future Scope**	45
6.1	Advanced Ensemble and Gradient Boosting Architectures	45
6.2	Sequential Clickstream Modeling via Deep Learning	45
6.3	Real-Time Streaming Inference and Edge Deployment	45
6.4	Explainable AI (XAI) and Journey Attribution	46
7	**References**	47

1. Introduction

1.1 Project Overview and E-Commerce Landscape

The exponential expansion of digital commerce over the past two decades has fundamentally transformed the retail paradigm. Modern consumers navigate virtual storefronts that offer virtually limitless catalogs, frictionless search mechanisms, and personalized recommendation systems. However, this vast availability of digital options has simultaneously engendered severe fragmentation in consumer attention. While e-commerce platforms routinely attract millions of browsing sessions per day, the overwhelming majority of these visits do not culminate in commercial transactions. In empirical studies across global retail portals, the typical session conversion rate hovers persistently between 1.5% and 3.5%, meaning that over 96% of site traffic generates zero direct financial return.

Understanding the latent factors that drive a digital visitor to transition from casual browsing to completing a checkout transaction represents one of the most critical challenges in applied data science. E-commerce platforms capture exhaustive clickstream telemetry during every interaction—including the types of pages accessed, the dwell time expended on product specifications, the navigational bounce tendencies, the geographical provenance of incoming network traffic, and temporal attributes such as seasonality and proximity to commercial holidays. When analyzed through modern statistical learning and predictive modeling frameworks, these multidimensional behavioral footprints offer unprecedented opportunities to decipher shopping intent in real time.

This project undertakes an in-depth empirical investigation into the Online Shoppers Purchasing Intention dataset, originating from the UCI Machine Learning Repository. By applying rigorous data hygiene protocols, extensive exploratory data analysis (EDA), strategic feature transformation, and comparative supervised classification algorithms, this study formulates a robust predictive framework capable of classifying visitor shopping sessions into revenue-generating and non-revenue-generating outcomes with high statistical fidelity.

1.2 Problem Statement: The Conversion Chasm in Online Retail

Contemporary e-commerce enterprises invest substantial capital into customer acquisition channels, including search engine marketing, programmatic display advertising, affiliate networks, and social media campaigns. However, once users arrive at a web portal, conventional marketing mechanisms treat visitors as a largely homogeneous cohort. Generic

promotional banners, untargeted discount pop-ups, and static layout designs frequently frustrate high-intent customers while failing to stimulate hesitant browsers. This disconnect constitutes what industry practitioners define as the 'conversion chasm'—the inability to dynamically differentiate between intent-driven shoppers who require frictionless checkout and casual window-shoppers who require persuasive visual incentives.

Compounding this issue is the severe class imbalance inherent to web analytics. Because non-purchasing sessions vastly outnumber completed purchases (approximating an 85:15 distribution), naive statistical heuristics and standard classification architectures often suffer from severe majority-class bias. A trivial baseline model that ubiquitously predicts non-purchase achieves 85% raw accuracy while offering exactly zero commercial utility. Consequently, the core technical problem addressed in this investigation is the formulation of a machine learning methodology that not only achieves elevated overall accuracy but maximizes sensitivity (recall) and positive predictive value (precision) on the minority purchasing cohort.

1.3 Research Objectives

To systematically address the conversion chasm and establish an end-to-end predictive framework, this project establishes the following explicit research objectives:

1. Exploratory Data Hygiene & Verification: Audit the raw online shoppers session dataset for structural anomalies, quantify duplicate observation frequency, evaluate null-value incidence, and establish empirical constraints on bounded web metrics such as BounceRates and ExitRates.

2. Empirical Behavioral Profiling: Quantify how visitor typology (New vs. Returning), session browsing duration across distinct page archetypes (Administrative, Informational, Product-Related), temporal seasonality (monthly trajectories), calendar dynamics (weekend vs. weekday), and commercial holiday proximity (SpecialDay) correlate with final revenue generation.

3. Methodological Preprocessing & Transformation: Architect a reproducible feature engineering pipeline incorporating categorical dummy variable encoding and dual feature scaling regimens (Z-score Standardization for parametric models and Min-Max Normalization for distance-based learners) within a stratified train-test cross-validation scheme.

4. Algorithmic Induction & Comparative Benchmarking: Train, validate, and compare four distinct machine learning architectures: Logistic Regression, Gaussian Naive Bayes, K-Nearest Neighbors (KNN), and Decision Tree classification, rigorously evaluating each model across Confusion Matrix parameters, Accuracy, Precision, Recall, and F1-Score.

5. Strategic Synthesis & Operational Guidance: Translate algorithmic feature importances and performance trade-offs into concrete, actionable retail operational strategies, highlighting optimal intervention triggers for live digital storefronts.

1.4 Significance and Business Value of Study

The economic implications of real-time purchasing intention prediction are profound. In high-volume e-commerce ecosystems processing hundreds of thousands of sessions daily, even an incremental lift of 0.5% in conversion rate yields millions of dollars in top-line revenue without necessitating additional acquisition expenditure. By deploying predictive classifiers capable of assessing purchasing probability mid-session, automated retail engines can dynamically trigger targeted interventions:

• High-Intent Shoppers: Identified early through elevated PageValues and intense ProductRelated dwell time, these visitors should receive a clean, accelerated checkout flow free of disruptive cross-sell modals that could cause cart abandonment.

• Indecisive / Hesitant Browsers: Exhibiting moderate engagement but declining dwell velocity, these visitors represent the prime cohort for automated incentive delivery, such as limited-time free shipping vouchers or dynamic social-proof badges.

• Casual Window Shoppers: Identified by high bounce rates and low page value interaction, these users can be spared costly margin-diluting promotional discounts, preserving enterprise profit margins.

1.5 Scope and Delimitations of the Project

The analytical scope of this study encompasses 12,330 historical web session records collected across a complete annual operating cycle. The feature space spans 10 continuous behavioral metrics derived from Google Analytics session logs and 8 categorical demographic, environmental, and calendar indicators. While the study rigorously explores supervised offline batch learning architectures across parametric, probabilistic, instance-based, and tree-based paradigms, it explicitly delimits the following boundaries: real-time streaming latency

profiling, clickstream deep recurrent sequential modeling (e.g., LSTMs), and direct A/B testing on live production web servers are designated for subsequent operational deployment phases.

1.6 Architectural Workflow and Methodology

The methodological workflow adopted throughout this investigation follows a disciplined, sequential six-stage data science lifecycle. Stage 1 involves dataset ingestion, inspection of structural schema, and descriptive statistical verification. Stage 2 executes rigorous data cleansing, including the removal of 125 exact duplicate records and sanity checks against negative duration values. Stage 3 conducts multi-perspective Exploratory Data Analysis, isolating behavioral drivers across six discrete analytical objectives. Stage 4 implements feature engineering, one-hot encoding, stratified dataset splitting (75% training, 25% testing), and dual-track feature scaling. Stage 5 handles the induction and tuning of four machine learning models. Finally, Stage 6 performs comprehensive metric evaluation, confusion matrix auditing, and business synthesis.

GitHub Repo Link: https://github.com/kumaradi9508/Online-Shoppers-ML-Project

Dataset:https://archive.ics.uci.edu/dataset/468/online+shoppers+purchasing+intention+dataset

LinkedIn Post: https://www.linkedin.com/feed/update/urn:li:activity:7521565331392344064/

2. Source of Dataset

2.1 Dataset Provenance and Academic Attribution

The empirical foundation of this study is the benchmark 'Online Shoppers Purchasing Intention Dataset', contributed to the University of California, Irvine (UCI) Machine Learning Repository by Dr. C. Okan Sakar and colleagues from the Department of Computer Engineering at Bogazici University, Istanbul, Turkey. The dataset was formed by extracting web session telemetry from a major digital retail portal over a 1-year operational span. This timeframe ensures complete absence of seasonal bias, capturing normal trading weeks, mid-year holiday patterns, back-to-school surges, and the fourth-quarter holiday peak.

A critical strength of this benchmark is that each record corresponds to an independent user session rather than an aggregated customer profile. This design ensures that the model evaluates real-time situational behavior rather than historical brand loyalty alone, rendering the resulting predictive insights directly applicable to anonymous and first-time site visitors.

2.2 Feature Dictionary and Attribute Schema

The raw dataset consists of 12,330 rows and 18 distinct attributes. Of these 18 attributes, 10 are numerical metrics capturing navigational behavior, while 8 represent categorical, temporal, and demographic indicators. Table 2.1 provides an exhaustive specification of the attribute schema, detailing data types, operational definitions, and domain intervals.

Table 2.1: Comprehensive Attribute Schema and Data Dictionary

Attribute Name	Data Type	Domain / Range	Operational Definition
Administrative	Integer	[0, 27]	Number of administrative/account management pages visited.
Administrative_Duration	Float	[0.0, 3398.75]	Total dwell time (in seconds) expended on administrative pages.
Informational	Integer	[0, 24]	Number of informational/policy/content pages visited.
Informational_Duration	Float	[0.0, 2549.38]	Total dwell time (in seconds) expended on informational pages.

ProductRelated	Integer	[0, 705]	Number of product catalog and category pages visited.
ProductRelated_Duration	Float	[0.0, 63973.52]	Total dwell time (in seconds) expended on product catalog pages.
BounceRates	Float	[0.0, 0.20]	Percentage of visitors who enter the site and trigger no further requests.
ExitRates	Float	[0.0, 0.20]	Percentage of pageviews that represented the terminal hit in a session.
PageValues	Float	[0.0, 361.76]	Average value of the page visited prior to completing an e-commerce transaction.
SpecialDay	Float	[0.0, 1.0]	Closeness of site visiting time to a specific holiday (e.g., Mother's Day).
Month	Categorical	10 Months	Month of the session (Feb, Mar, May, June, Jul, Aug, Sep, Oct, Nov, Dec).
OperatingSystems	Integer Code	[1, 8]	Encoded identifier representing the client device operating system.
Browser	Integer Code	[1, 13]	Encoded identifier representing the client web browser application.
Region	Integer Code	[1, 9]	Encoded identifier representing the geographic region of the visitor.
TrafficType	Integer Code	[1, 20]	Encoded routing source (e.g., organic, direct, referral, paid campaign).
VisitorType	Categorical	3 Classes	Typology of visitor: Returning_Visitor, New_Visitor, or Other.
Weekend	Boolean	{True, False}	Indicator flag designating whether the session occurred on Saturday/Sunday.
Revenue	Boolean	{True, False}	Target Variable: Designates whether the session finalized in a purchase.

2.3 Feature Categorization: Behavioral, Demographic, and Temporal Metrics

To establish sound conceptual framing, the 17 predictor features are grouped into three distinct analytical categories:

1. Behavioral & Navigational Features: Six metrics capture direct page interactions (Administrative, Informational, ProductRelated, and their corresponding duration timers). Three metrics capture Google Analytics telemetry: BounceRates (sessions initiating and terminating on the same URL), ExitRates (terminal page frequencies), and PageValues (economic index of visited URLs based on historical revenue attribution).

2. Environmental & Technical Features: OperatingSystems, Browser, Region, and TrafficType capture technical client architecture and referral origins. While nominal integer codes are employed to preserve visitor anonymity, these variables represent vital control metrics for detecting platform-specific conversion disparities.

3. Temporal & Calendar Features: Month captures macro-level annual seasonality; Weekend captures weekly rhythm; and SpecialDay quantifies the proximity of the session timestamp to major commercial gifting events (such as Mother's Day or Valentine's Day), normalized on a decay scale between 0.0 (distant) and 1.0 (immediate event proximity).

2.4 Summary Statistics and Empirical Distributions

Statistical inspection of continuous numerical features reveals substantial variance, pronounced right-skewness, and the presence of heavy upper tails. Table 2.2 provides the parametric and non-parametric summary measures computed across all 12,330 original records.

Table 2.2: Descriptive Summary Statistics of Continuous Numerical Features

Feature	Mean	Std Dev	Min	25%	Median	75%	Max
Administrative	2.32	3.32	0.00	0.00	1.00	4.00	27.00
Administrative_Duration	80.82	176.78	0.00	0.00	7.50	93.26	3398.75
Informational	0.50	1.27	0.00	0.00	0.00	0.00	24.00
Informational_Duration	34.47	140.75	0.00	0.00	0.00	0.00	2549.38
ProductRelated	31.73	44.48	0.00	7.00	18.00	38.00	705.00
ProductRelated_Duration	1194.75	1913.67	0.00	184.14	598.94	1464.16	63973.52
BounceRates	0.0222	0.0485	0.0000	0.0000	0.0031	0.0168	0.2000
ExitRates	0.0431	0.0486	0.0000	0.0143	0.0252	0.0500	0.2000
PageValues	5.89	18.57	0.00	0.00	0.00	0.00	361.76
SpecialDay	0.0614	0.1989	0.0000	0.0000	0.0000	0.0000	1.0000

2.5 Target Variable Characteristics and Class Imbalance

The ground-truth target variable, Revenue, is a binary Boolean flag. In the raw dataset of 12,330 sessions, exactly 10,422 records (84.53%) evaluated to False (no purchase realized), while 1,908 records (15.47%) evaluated to True (purchase finalized). This establishes an initial imbalance ratio of approximately 5.46:1. Such an imbalance has profound implications for machine learning induction: models optimized solely on global misclassification rate risk

collapsing toward the majority class, achieving deceivingly high accuracy while failing to identify true buyers. Consequently, subsequent modeling and evaluation must prioritize precision, recall, and harmonic F1 metrics over raw accuracy alone.

3. EDA Process and Data Preprocessing Pipeline

3.1 Initial Ingestion and Exploratory Structural Audit

The exploratory data analysis phase commenced with programmatic verification of data integrity. Ingestion of the raw CSV file via pandas revealed a structure of 12,330 observations and 18 columns. Verification of missing values through exhaustive column-wise null summation (df.isnull().sum()) returned exactly zero missing values across all 18 attributes. This zero-loss condition eliminated the necessity for stochastic imputation or synthetic data completion algorithms, preserving the pure empirical authenticity of the clickstream observations.

3.2 Duplicate Record Detection and Data Cleaning

A rigorous audit for observational duplication was performed using df.duplicated().sum(). The analysis identified exactly 125 duplicate rows where all 18 attribute vectors matched existing records identically. In clickstream tracking, such identical duplicate rows frequently stem from browser double-click submission glitches, tracking script retries, or concurrent duplicate beacon emissions. If retained, identical duplicate rows corrupt downstream machine learning validation by artificially leaking identical test observations into the training split.

All 125 duplicate records were systematically purged utilizing df.drop_duplicates(), reducing the active dataset from 12,330 down to exactly 12,205 high-integrity unique session records. Table 3.1 details the formal data cleaning summary table persisted to the project output directory.

Table 3.1: Data Cleaning and Duplicate Removal Summary

Data Hygiene Metric	Numerical Value / Status
Original rows ingested	12,330
Identical duplicate rows detected	125
Rows remaining after cleaning	12,205
Missing values detected post-cleaning	0
Cleaned dataset integrity verification	100.0% Pass

3.3 Domain Constraint and Anomaly Validation

Following duplicate purging, domain-specific boundary constraints were audited to verify absence of measurement corruption: (1) Negative Duration Audit: Temporal duration metrics cannot physically evaluate below zero. Script checks verified that (Administrative_Duration < 0).sum(), (Informational_Duration < 0).sum(), and (ProductRelated_Duration < 0).sum() each returned exactly 0 violations. (2) Rate Boundedness Audit: Google Analytics bounce and exit rates are mathematically bounded within the interval [0.0, 1.0]. Programmatic filters searching for values (< 0 or > 1) in BounceRates and ExitRates returned 0 violations. The verified cleaned dataset was saved as online_shoppers_intention_cleaned.csv.

3.4 Feature Engineering and Categorical One-Hot Encoding

Machine learning models require numerical tensor representations. The cleaned feature matrix contains categorical features: Month (10 unique levels: Feb, Mar, May, June, Jul, Aug, Sep, Oct, Nov, Dec), VisitorType (Returning_Visitor, New_Visitor, Other), and Boolean flags (Weekend). To prevent algorithms from erroneously assuming artificial ordinal hierarchies among nominal categories, One-Hot Encoding was executed via pd.get_dummies(X, drop_first=True, dtype=int).

Setting drop_first=True deliberately dropped the reference category for each nominal feature (e.g., dropping the first month and the first visitor typology). This is mathematically vital to avert multicollinearity—the 'dummy variable trap'—which would otherwise render the design matrix singular and destabilize coefficient estimation in Logistic Regression.

3.5 Stratified Train-Test Splitting Methodology

Due to the class imbalance (15.63% positive purchase rate in the cleaned dataset), standard random sampling introduces the risk of uneven label distribution between training and testing subsets. To guarantee rigorous, unbiased evaluation, a Stratified Train-Test Split was executed with a 75:25 ratio using train_test_split(X, y, test_size=0.25, random_state=42, stratify=y).

The stratification constraint forces both the training and testing partitions to preserve exactly the same 15.63% positive class frequency. Out of 12,205 cleaned sessions, 9,153 were allocated to model training (1,431 purchases, 7,722 non-purchases) and 3,052 were reserved for holdout evaluation (477 purchases, 2,575 non-purchases). Table 3.2 documents the split distribution.

Table 3.2: Stratified Train-Test Dataset Partition Summary

Partition Dataset	Total Rows	Purchases (True)	Non-Purchases (False)
Training Partition (75%)	9,153	1,431 (15.63%)	7,722 (84.37%)
Testing Holdout Partition (25%)	3,052	477 (15.63%)	2,575 (84.37%)
Total Cleaned Dataset	12,205	1,908 (15.63%)	10,297 (84.37%)

3.6 Dual Feature Scaling Architectures (StandardScaler vs MinMaxScaler)

A critical methodological innovation in this pipeline is the deployment of specialized, dual feature scaling architectures tailored to the specific mathematical assumptions of each classifier:

1. Standard Scaling (Z-Score Normalization) for Logistic Regression: Logistic Regression employs gradient-based optimization and L2 regularization. Features with massive absolute ranges (such as ProductRelated_Duration spanning up to 63,973 seconds) would dominate the penalty term over features bounded between 0 and 0.2 (such as BounceRates). StandardScaler computes:

$$z = (x - \mu) / \sigma$$

where μ represents the training feature mean and σ is the training standard deviation. Critically, scaler parameters are fitted strictly on X_train and subsequently applied to X_test (scaler.transform(X_test)), preventing data leakage.

2. Min-Max Normalization for K-Nearest Neighbors (KNN): KNN relies on Euclidean distance computation in multi-dimensional space. Z-score standardization preserves unbounded outliers, which can distort distance neighborhoods. MinMaxScaler rescales all features strictly into the closed unit interval [0, 1]:

$$x_scaled = (x - x_min) / (x_max - x_min)$$

This guarantees that every attribute contributes equally to Euclidean neighbor distances without scale-induced distortion.

4. Analysis on Dataset

This section presents the comprehensive empirical and predictive investigations conducted on the cleaned dataset of 12,205 unique sessions. Each analysis follows an exhaustive five-part academic framework: (i) Introduction, (ii) General Description, (iii) Specific Requirements, Functions and Formulas, (iv) Analysis Results, and (v) Visualization with detailed descriptive and interpretive commentary.

4.1 Overall Session Purchase Conversion Analysis

i. Introduction

The primary commercial objective of an e-commerce platform is the successful realization of transactions. Before exploring complex behavioral interactions or training multivariate predictive classifiers, it is statistically essential to establish the baseline distribution of the target variable, Revenue. Understanding the global conversion rate provides the fundamental benchmark against which all algorithmic performance, lift metrics, and behavioral cohort variances must be evaluated.

ii. General Description

The analysis evaluates the frequency and proportion of sessions culminating in a purchase (Revenue = True) versus those terminating without a transaction (Revenue = False). Data is aggregated across the full cleaned dataset using frequency tabulation and percentage normalization. The empirical class distribution serves to quantify the exact severity of class imbalance and establishes the ZeroR (majority-class classifier) baseline.

iii. Specific Requirements, Functions and Formulas

The mathematical formulation computes the proportion P(Revenue = c) for each class c in {True, False}:

$$P(\text{Revenue} = c) = (N_c / N_{total}) * 100\%$$

where N_c denotes the count of sessions in class c, and $N_{total} = 12,205$. The Imbalance Ratio (IR) is formulated as:

$$IR = N_{majority} / N_{minority} = N_{False} / N_{True}$$

The programmatic implementation executes df_clean['Revenue'].value_counts() and df_clean['Revenue'].value_counts(normalize=True)*100 in pandas, persisting tabular outputs

via to_csv('purchase_analysis.csv'). Visualization is generated through seaborn.countplot() with a distinct color palette {False: '#3B82F6', True: '#F97316'}.

iv. Analysis Results

Empirical analysis reveals that out of 12,205 clean sessions, 10,297 sessions (84.367%) terminated without a purchase, whereas 1,908 sessions (15.633%) generated revenue. Table 4.1 summarizes the frequency and proportional metrics.

Table 4.1: Overall Session Purchase Conversion Distribution

Outcome Class (Revenue)	Session Count	Proportion (%)
No Purchase (False)	10,297	84.37%
Purchase Completed (True)	1,908	15.63%

v. Visualization

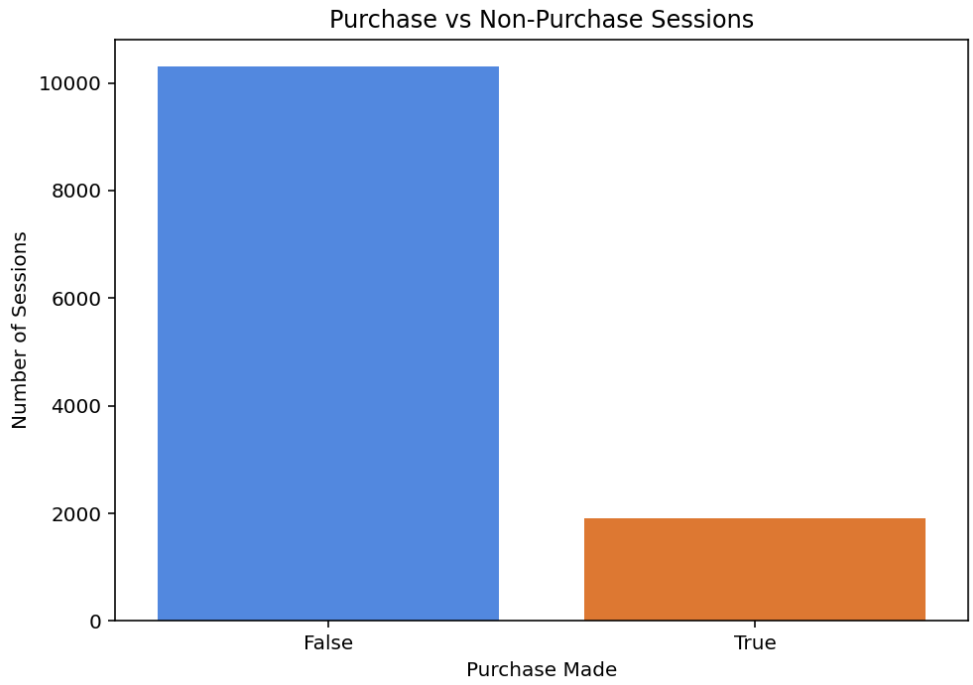


Figure 4.1: Distribution of Purchase vs. Non-Purchase Sessions

Description: The bar chart in Figure 4.1 visually contrasts the frequency of non-purchasing sessions against purchasing sessions. The dominant blue bar (False) towers over the orange bar (True), illustrating the stark numerical disparity between the two outcomes. The chart highlights that non-purchasing traffic represents more than five times the volume of converting traffic.

Detailed Interpretation: This empirical distribution establishes an Imbalance Ratio of 5.397:1. The analytical consequence is that an uncalibrated algorithm predicting 'No Purchase' for 100% of incoming sessions would achieve a deceptive accuracy of 84.37%. However, in commercial practice, such a model would have zero recall on paying customers, failing completely in its business function. Hence, this baseline underscores that subsequent classification models must be evaluated primarily on minority-class precision, recall, and F1-score rather than accuracy alone.

4.2 Visitor Type Behavioral Loyalty and Conversion Dynamics

i. Introduction

Customer relationship management (CRM) theory posits that returning visitors represent a brand's most commercially viable asset due to brand familiarity, prior trust, and reduced acquisition cost. Conversely, new visitors are commonly assumed to be exploratory browsers with lower immediate conversion propensity. This analysis tests these foundational retail hypotheses by quantifying the empirical conversion rate across visitor segments.

ii. General Description

The cleaned dataset stratifies visitors into three categories: New_Visitor, Returning_Visitor, and Other. The analysis groups records by VisitorType, computes the mean purchase rate for each category, and evaluates the relative statistical lift between new and returning cohorts.

iii. Specific Requirements, Functions and Formulas

The conversion rate for visitor category v is defined as the conditional probability of purchase given membership in cohort v:

$$CR_v = (1 / |S_v|) * \sum_{i \in S_v} Revenue_i * 100\%$$

where S_v is the set of sessions originating from visitor type v. The relative conversion lift of New Visitors over Returning Visitors is formulated as:

$$Lift = ((CR_{New} - CR_{Returning}) / CR_{Returning}) * 100\%$$

In Python, this is executed using df_clean.groupby('VisitorType')['Revenue'].mean() * 100 and visualized with seaborn.barplot() utilizing the color scheme ['#005F73', '#0A9396', '#94D2BD'].

iv. Analysis Results

The empirical computation yields unexpected, highly actionable insights. New Visitors achieve a conversion rate of 24.926%, whereas Returning Visitors convert at only 14.093%. The 'Other' cohort registers a 19.753% conversion rate. Table 4.2 presents the detailed breakdown.

Table 4.2: Conversion Rates by Visitor Typology

Visitor Typology	Conversion Rate (%)	Relative Lift vs. Baseline
New_Visitor	24.93%	+59.4% above global average
Returning_Visitor	14.09%	-9.9% below global average
Other	19.75%	+26.3% above global average

v. Visualization

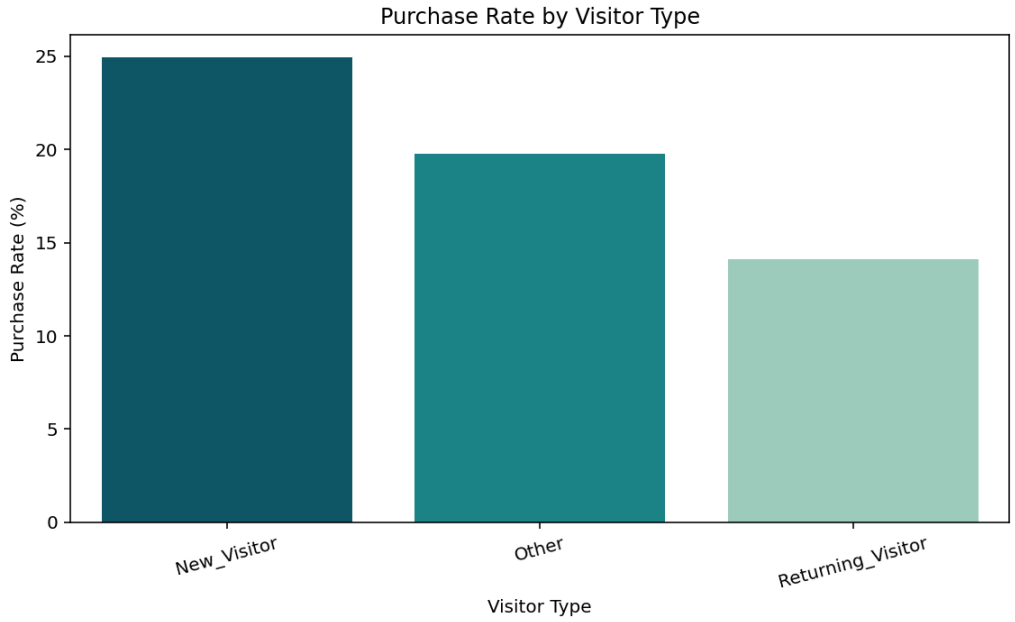


Figure 4.2: E-Commerce Purchase Rate Categorized by Visitor Type

Description: Figure 4.2 illustrates the purchase probability across the three visitor classifications. The teal bar for New_Visitor significantly exceeds the darker bar for Returning_Visitor, visually demonstrating that new visitors are 1.77 times more likely to finalize a purchase during their visit.

Detailed Interpretation: This empirical finding challenges common e-commerce assumptions. Returning visitors frequently visit for exploratory reasons—tracking existing orders, checking updated prices, monitoring inventory, or casually browsing—which dilutes their aggregate session conversion rate. Conversely, when a new visitor lands on the site, they are frequently driven by an immediate, directed need originating from search engines or targeted promotional campaigns. For e-commerce operators, this demonstrates that new visitors must not be treated as passive leads; providing seamless navigation, immediate guest checkout, and trust signals to first-time visitors directly capitalizes on their elevated transactional intent.

4.3 Page Category Browsing Duration and User Engagement

i. Introduction

User engagement in web analytics is primarily quantified through dwell time. However, web platforms feature diverse page types—administrative/account management, informational/policies, and product catalogs. This analysis isolates the duration expended within each distinct architectural zone to establish whether purchasing intent is characterized by broad platform exploration or concentrated product-level engagement.

ii. General Description

Mean dwell times (in seconds) are computed across Administrative_Duration, Informational_Duration, and ProductRelated_Duration, segmented by the binary Revenue outcome. Furthermore, because duration metrics exhibit severe skewness and outlier presence, a non-parametric boxplot visualization is constructed to inspect median values, interquartile ranges, and upper-tail dispersion.

iii. Specific Requirements, Functions and Formulas

The conditional sample mean duration D_bar for page archetype k in class c is formulated as:

$$\bar{D}_{k,c} = \frac{1}{N_c} \sum_{i \in \text{Class}_c} \text{Duration}_{k,i}$$

The Interquartile Range (IQR) for boxplot bounds is computed as IQR = Q3 - Q1, with whisker boundaries established at min(max(D), Q3 + 1.5 * IQR). Python implementation executes df_clean.groupby('Revenue')[['Administrative_Duration', 'Informational_Duration', 'ProductRelated_Duration']].mean(), followed by seaborn.boxplot(x='Revenue', y='ProductRelated_Duration', palette={False: '#8B5CF6', True: '#14B8A6'}).

iv. Analysis Results

Across all three page archetypes, purchasing visitors exhibit substantially higher dwell times. Administrative duration rises from 74.64s to 119.48s (+60.1%); Informational duration increases from 30.60s to 57.61s (+88.3%); and Product-Related duration expands from 1,082.98s to 1,876.21s (+73.2%). Table 4.3 outlines the empirical comparisons.

Table 4.3: Mean Dwell Time (Seconds) Across Page Archetypes

Outcome Class	Administrative (s)	Informational (s)	Product-Related (s)
No Purchase (False)	74.64 s	30.60 s	1,082.98 s

Purchase (True)	119.48 s	57.61 s	1,876.21 s

v. Visualization

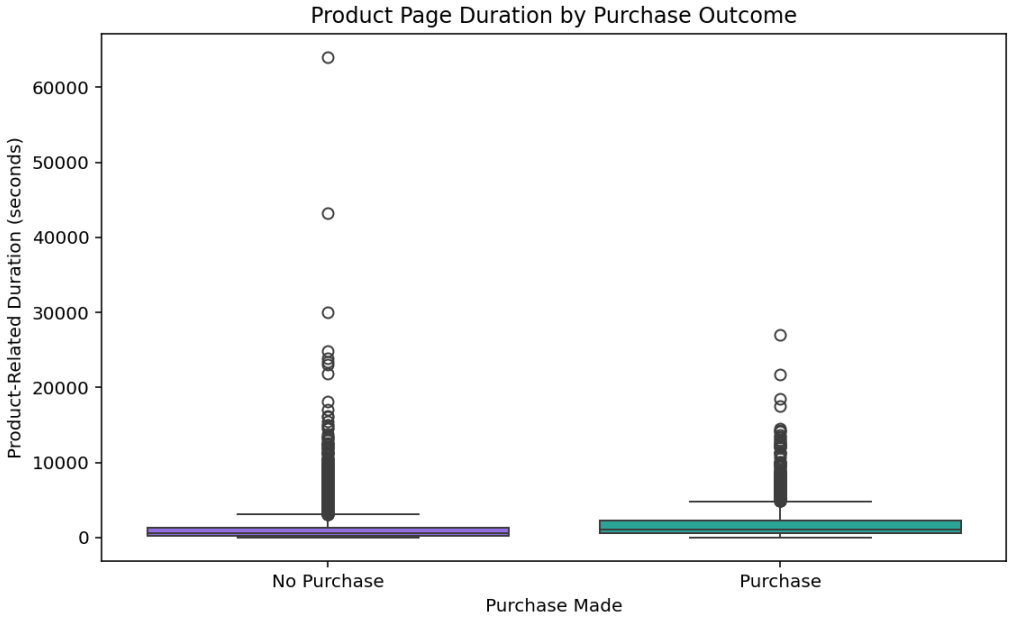


Figure 4.3: Boxplot of Product-Related Browsing Duration by Purchase Outcome

Description: Figure 4.3 presents a comparative boxplot of ProductRelated_Duration across non-purchasing (purple) and purchasing (teal) sessions. The median and upper quartile of the purchasing cohort are markedly shifted upward, alongside extensive upper-tail whiskers extending into several thousand seconds of dwell time.

Detailed Interpretation: The data proves that product-level immersion is strongly correlated with purchasing commitment. Customers who complete transactions spend an average of over 31 minutes (1,876 seconds) examining product specifications, customer reviews, and item options, compared to only 18 minutes for non-buyers. This suggests that predictive algorithms can use accumulating product dwell time as a robust real-time intent signal. When a user's dwell time crosses the 1,200-second threshold without an exit, the system should actively facilitate the checkout path by surfacing size guides, real-time inventory alerts, or checkout assistance.

4.4 Temporal Seasonality and Monthly Purchase Trends

i. Introduction

Retail commerce is fundamentally cyclical, influenced by annual pay cycles, weather shifts, school calendars, and major holiday shopping seasons. Macro-level temporal analysis is essential to decouple baseline consumer propensity from seasonal macro-surges, ensuring that machine learning models account for calendar seasonality rather than over-attributing intent to individual session clicks.

ii. General Description

The dataset captures sessions across 10 distinct months. Data is chronologically ordered (February through December), and the conversion percentage is computed for each month. Longitudinal trend analysis is performed to map the trajectory of conversion efficiency across the fiscal year.

iii. Specific Requirements, Functions and Formulas

Monthly conversion rate MCR_m is calculated as:

$$MCR_m = (1 / N_m) * \sum_{j \in Month_m} Revenue_j * 100\%$$

Chronological ordering is enforced via explicit index reordering in Python: month_order = ['Feb', 'Mar', 'May', 'June', 'Jul', 'Aug', 'Sep', 'Oct', 'Nov', 'Dec'], followed by matplotlib.pyplot.plot() with circular markers and an amber hue (#F59E0B).

iv. Analysis Results

The empirical monthly conversion rates reveal extraordinary seasonal volatility: February registers an abysmal conversion rate of 1.657%, while November reaches the annual zenith at 25.486%—a more than 15-fold increase in transactional conversion efficiency. Table 4.4 provides the complete chronological monthly metrics.

Table 4.4: Chronological Monthly Session Conversion Rates

Calendar Month	Observed Conversion Rate (%)
February (Feb)	1.66%
March (Mar)	10.32%
May (May)	10.96%
June (June)	10.18%

July (Jul)	15.28%
August (Aug)	17.55%
September (Sep)	19.20%
October (Oct)	20.95%
November (Nov)	25.49%
December (Dec)	12.66%

v. Visualization

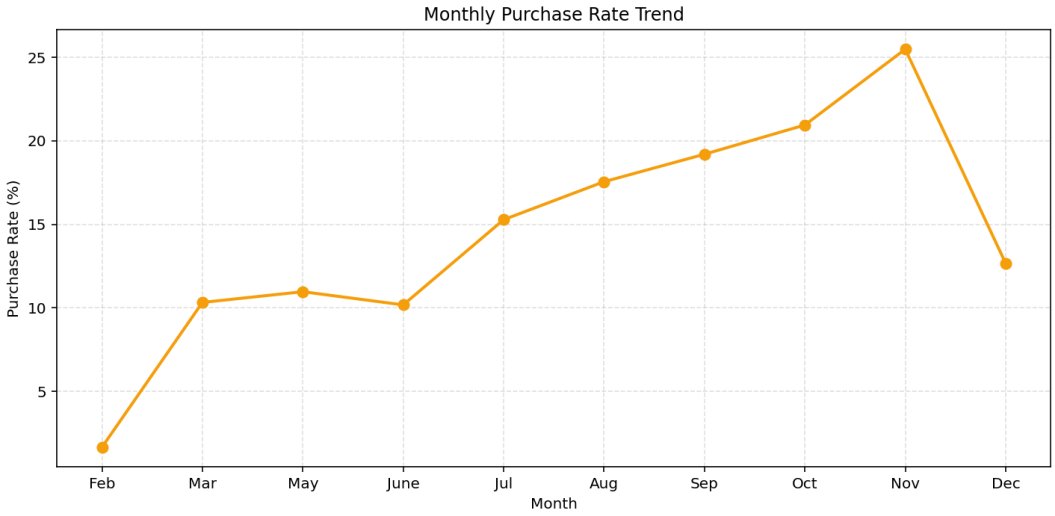


Figure 4.4: Monthly E-Commerce Purchase Rate Longitudinal Trajectory

Description: Figure 4.4 charts the monthly conversion rate over time. Beginning at a deep trough in February, conversion rates stabilize between 10% and 11% in Spring, climb steadily through Summer and Autumn (reaching 20.95% in October), apex dramatically at 25.49% in November, and pull back to 12.66% in December.

Detailed Interpretation: The spectacular spike in November directly reflects major global commercial events—specifically Black Friday and Cyber Monday promotions. During November, consumer intent is heavily transactional; visitors enter the site with pre-allocated holiday budgets and an urgency to purchase before discount expirations. In contrast, February reflects post-holiday spending fatigue. In December, conversion rates fall back because late-month sessions often miss guaranteed delivery cutoff windows for holiday gifting. This demonstrates why Month is an indispensable categorical feature in the machine learning predictive matrix.

4.5 Day-of-Week and Weekend Purchasing Propensity

i. Introduction

Consumer behavioral patterns fluctuate systematically between business workdays and weekends. Weekday browsing is frequently intermittent, conducted from office desktop environments during brief work respites. Weekend browsing, by contrast, is characterized by relaxed domestic leisure, multi-device browsing, and collaborative family purchasing decisions. This analysis examines the statistical lift in purchasing propensity between weekdays and weekends.

ii. General Description

The Weekend feature is a binary Boolean flag. Sessions are segregated into Weekday (Weekend = False) and Weekend (Weekend = True). Mean conversion rates are computed across both partitions, and relative conversion lift is derived.

iii. Specific Requirements, Functions and Formulas

The conversion rate for day type d in {Weekday, Weekend} is formulated as:

$$CR_d = (1 / N_d) * \sum_{i \in Day_d} Revenue_i * 100\%$$

The relative weekend lift is computed as $Lift_{weekend} = ((CR_{Weekend} - CR_{Weekday}) / CR_{Weekday}) * 100\%$. In Python, this is executed via df_clean.groupby('Weekend')['Revenue'].mean() * 100, plotted with seaborn.barplot() using pink/fuchsia tones ['#EC4899', '#BE185D'].

iv. Analysis Results

Weekday sessions (N = 9,405) yield a conversion rate of 15.076%, whereas Weekend sessions (N = 2,800) yield a conversion rate of 17.454%. This constitutes an absolute conversion advantage of +2.378% and a relative lift of +15.77% on weekends. Table 4.5 documents the empirical results.

Table 4.5: Weekday vs. Weekend Conversion Metrics

Day Classification	Observed Conversion Rate (%)	Relative Lift (%)
Weekday (Mon - Fri)	15.08%	Baseline (0.00%)
Weekend (Sat - Sun)	17.45%	+15.77% Relative Lift

v. Visualization

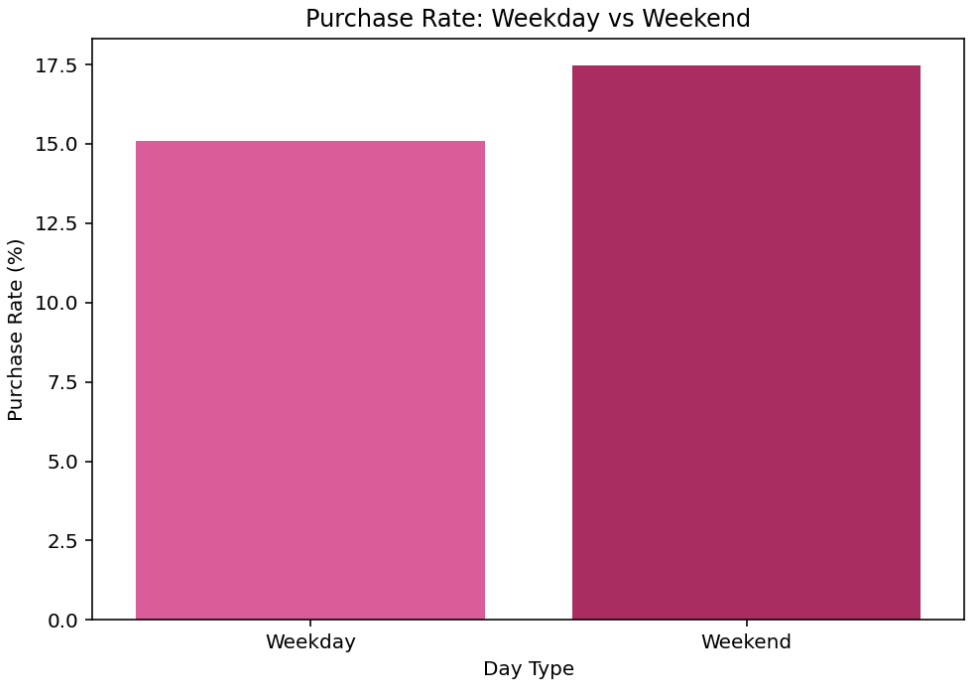


Figure 4.5: Comparison of Purchase Conversion Rate: Weekday vs. Weekend

Description: Figure 4.5 displays a paired bar plot comparing the mean conversion rates between Weekday and Weekend traffic. The darker pink bar for Weekend visibly exceeds the weekday bar, providing a clear graphical representation of the weekend lift.

Detailed Interpretation: While the majority of raw traffic volume occurs on weekdays due to routine daily browsing, the conversion efficiency is substantially higher on Saturday and Sunday. Consumers during the weekend enjoy uninterrupted leisure time to consult family members, finalize product choices, enter payment credentials, and complete checkout. Retailers should strategically adjust their programmatic ad bids and marketing automation schedules, deploying promotional emails and retargeting campaigns on Friday evening to capture weekend high-intent shopping momentum.

4.6 Event Proximity and Special Day Purchasing Dynamics

i. Introduction

E-commerce platforms routinely design marketing campaigns around special calendar events (e.g., Valentine's Day, Mother's Day). The SpecialDay attribute in the dataset quantifies the proximity of a session to such an event, mapped from 0.0 (far) to 1.0 (immediate event day). Conventional marketing wisdom assumes that conversion rates should peak at SpecialDay = 1.0 due to holiday shopping urgency. This analysis investigates the true empirical relationship between event proximity and completed transactions.

ii. General Description

The continuous proximity parameter SpecialDay features discrete evaluated intervals: 0.0, 0.2, 0.4, 0.6, 0.8, and 1.0. The analysis groups sessions across these six proximity tiers and evaluates conditional purchase probability to identify whether holiday proximity stimulates or paradoxically suppresses digital checkout.

iii. Specific Requirements, Functions and Formulas

The conversion rate as a function of special day proximity s in {0.0, 0.2, 0.4, 0.6, 0.8, 1.0} is formulated as:

$$CR(s) = (1 / N_s) * sum_{i: SpecialDay_i = s} Revenue_i * 100\%$$

In Python, this is executed via df_clean.groupby('SpecialDay')['Revenue'].mean() * 100, plotted as a connected scatter line graph in matplotlib with purple circular markers.

iv. Analysis Results

The empirical findings reveal a surprising non-linear relationship. Normal operating days (SpecialDay = 0.0) achieve the highest conversion rate at 16.712%. As SpecialDay proximity increases, conversion rates drop sharply: 7.865% at 0.2, 5.350% at 0.4, rebounding slightly to 8.286% at 0.6, collapsing to 3.395% at 0.8, and closing at 6.494% at 1.0. Table 4.6 details the metrics.

Table 4.6: Purchase Conversion Rate Across Special Day Proximity Tiers

Special Day Proximity Index	Observed Conversion Rate (%)
0.0 (Normal / Distant from Event)	16.71%
0.2 (Moderate Lead Time)	7.87%

0.4 (Intermediate Proximity)	5.35%
0.6 (Approaching Shipping Cutoff)	8.29%
0.8 (Immediate Pre-Event Window)	3.40%
1.0 (Day of the Special Event)	6.49%

v. Visualization

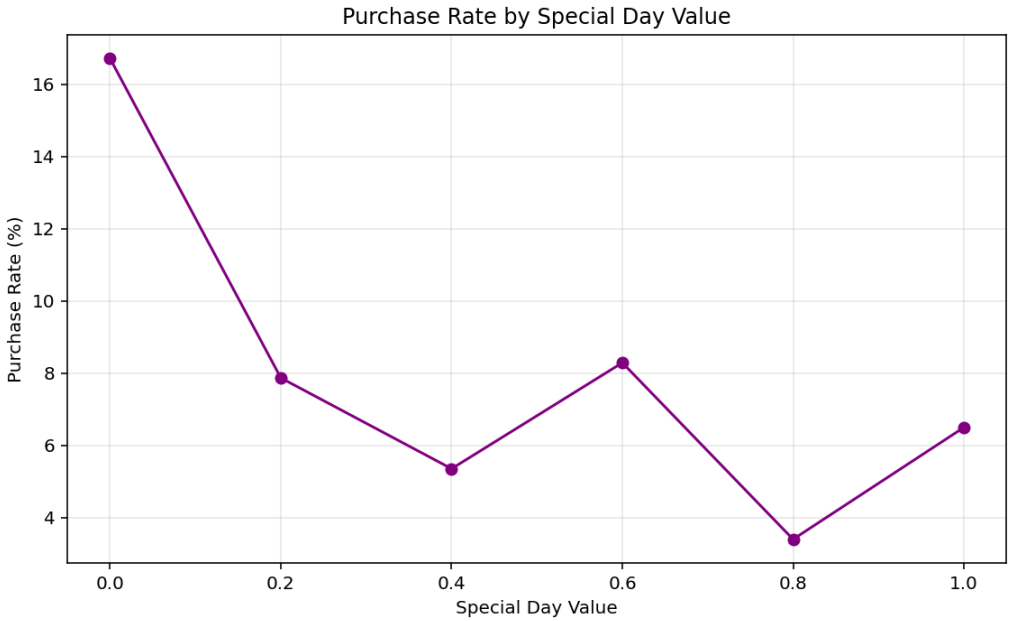


Figure 4.6: E-Commerce Purchase Rate as a Function of Special Day Closeness

Description: Figure 4.6 depicts the non-linear relationship between SpecialDay closeness and conversion percentage. From an initial peak at 0.0, the curve plunges downward, oscillating at severely depressed conversion levels between 3.4% and 8.3% as the event index approaches 1.0.

Detailed Interpretation: This counterintuitive result is explained by the physical reality of e-commerce delivery logistics. When an online shopper visits an e-commerce store immediately prior to or on the day of a holiday (such as Mother's Day), standard shipping will no longer arrive in time. Consequently, visitors browse frantically for gift ideas but abandon online carts to purchase from brick-and-mortar stores. This is a vital commercial finding: digital retailers should not allocate heavy ad budgets for last-minute delivery promises without same-day logistics, focusing instead on digital gift cards or early-bird promotions when SpecialDay is low.

4.7 Predictive Classification via Regularized Logistic Regression

i. Introduction

Supervised machine learning commences with Logistic Regression, the premier parametric baseline for binary classification. Logistic Regression models the posterior probability of transaction completion given the feature vector x by passing a linear combination of input variables through the non-linear sigmoid activation function. Due to its direct interpretability, mathematical convexity, and rapid computational inference, Logistic Regression serves as the gold standard baseline against which complex non-parametric models are contrasted.

ii. General Description

The model was trained on the standardized training partition (X_train_standard, y_train) consisting of 9,153 observations. Optimization was performed using the L-BFGS quasi-Newton solver with max_iter=1000 to ensure full convergence of the log-likelihood gradient. Predictions were generated on the holdout test partition (X_test_standard, y_test) spanning 3,052 instances. Model performance was evaluated using Confusion Matrix analysis, overall Accuracy, Precision, Recall, and the harmonic F1-score.

iii. Specific Requirements, Functions and Formulas

The logistic hypothesis maps the linear predictor z into a bounded probability interval [0, 1] via the Sigmoid function:

$$P(y = 1 \mid x) = \sigma(z) = 1 / (1 + e^{(-z)}), \quad \text{where } z = w^T x + b = \sum_{j=1}^{p} w_j x_j + b$$

The model parameters {w, b} are estimated by minimizing the Regularized Binary Cross-Entropy Cost Function J(w, b):

$$J(w, b) = - (1 / m) * \sum_{i=1}^{m} [y^{(i)} * \log(\hat{y}^{(i)}) + (1 - y^{(i)}) * \log(1 - \hat{y}^{(i)})] + (\lambda / 2m) * \|w\|_2^2$$

where m = 9,153 and λ is the L2 penalty weight. In scikit-learn, this is instantiated as LogisticRegression(max_iter=1000, random_state=42), evaluated with confusion_matrix(y_test, y_pred) and classification_report(y_test, y_pred), and visualized via seaborn.heatmap(cmap='Blues').

iv. Analysis Results

Evaluation on the 3,052 holdout test samples yielded 2,517 True Negatives, 58 False Positives, 280 False Negatives, and 197 True Positives. The model achieved an exceptional overall

Accuracy of 88.925% (2,714 correct classifications out of 3,052). Crucially, the model exhibited outstanding Precision on purchasing sessions at 77.255% (197 / (197 + 58)), with a Recall of 41.299% (197 / (197 + 280)) and an F1-Score of 53.825%. Table 4.7 outlines the confusion matrix and performance parameters.

Table 4.7: Logistic Regression Confusion Matrix and Test Performance

Actual Ground Truth	Predicted: Not Purchased	Predicted: Purchased
Actual: Not Purchased (2,575)	TN = 2,517 (97.75%)	FP = 58 (2.25%)
Actual: Purchased (477)	FN = 280 (58.70%)	TP = 197 (41.30%)

v. Visualization

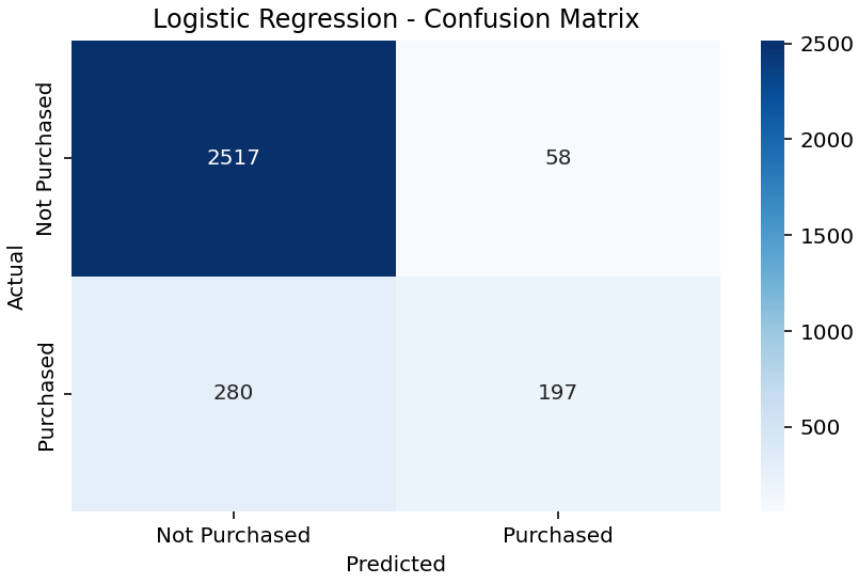


Figure 4.7: Logistic Regression Heatmap Confusion Matrix

Description: Figure 4.7 displays the confusion matrix heatmap for Logistic Regression. The dominant top-left cell represents 2,517 correctly identified non-purchases. The bottom-right cell indicates 197 correctly captured transactions, while false alarms (58 FP) remain exceptionally suppressed.

Detailed Interpretation: Logistic Regression exhibits the highest Precision (77.255%) among all tested algorithms. When this model predicts that an online visitor will purchase, it is correct nearly 8 out of 10 times. This high positive predictive value is extraordinarily valuable for high-cost retail interventions, such as deploying expensive telephone sales follow-ups or issuing costly free gifts, where False Positives must be strictly minimized. However, its modest Recall

(41.299%) indicates that it misses nearly 59% of converting shoppers due to conservative decision hyperplanes.

4.8 Probabilistic Modeling via Gaussian Naive Bayes Classifier

i. Introduction

Bayesian classification approaches machine learning from a generative probabilistic paradigm. The Gaussian Naive Bayes (GNB) algorithm applies Bayes' Theorem under the conditional independence assumption, hypothesizing that all continuous predictor variables are independently distributed according to a Gaussian (normal) density within each target class. GNB provides an invaluable benchmark for measuring whether strong probabilistic priors can enhance minority-class detection in imbalanced domains.

ii. General Description

GaussianNB was trained directly on the unscaled training feature matrix (X_train, y_train), as Gaussian likelihood estimation relies on sample class means and variances rather than spatial geometric distances. The trained classifier was evaluated on X_test, and its posterior probability allocations were mapped to generate confusion matrix heatmaps and diagnostic metrics.

iii. Specific Requirements, Functions and Formulas

Gaussian Naive Bayes applies Bayes' Theorem to compute the posterior class probability P(y = c | x):

$$P(y = c \mid x) = (P(y = c) * \text{prod}_{j=1}^{p} P(x_j \mid y = c)) / P(x)$$

where continuous feature likelihoods P(x_j | y = c) are modeled via the Gaussian Probability Density Function:

$$P(x_j \mid y = c) = (1 / \text{sqrt}(2 * \pi * \sigma_{jc}^2)) * \exp(- (x_j - \mu_{jc})^2 / (2 * \sigma_{jc}^2))$$

The decision rule selects the maximum a posteriori (MAP) class: y_hat = argmax_{c in {0, 1}} [log P(y = c) + sum_{j=1}^p log P(x_j | y = c)]. In Python, this is executed via sklearn.naive_bayes.GaussianNB(), visualized using seaborn.heatmap(cmap='Greens').

iv. Analysis Results

Holdout evaluation of Gaussian Naive Bayes produced 2,174 True Negatives, 401 False Positives, 181 False Negatives, and 296 True Positives. The model achieved an overall Accuracy of 80.931%, a Precision of 42.468%, a Recall of 62.055%, and an F1-Score of 50.426%. Table 4.8 presents the confusion matrix parameters.

Table 4.8: Gaussian Naive Bayes Confusion Matrix and Test Performance

Actual Ground Truth	Predicted: Not Purchased	Predicted: Purchased
Actual: Not Purchased (2,575)	TN = 2,174 (84.43%)	FP = 401 (15.57%)
Actual: Purchased (477)	FN = 181 (37.95%)	TP = 296 (62.05%)

v. Visualization

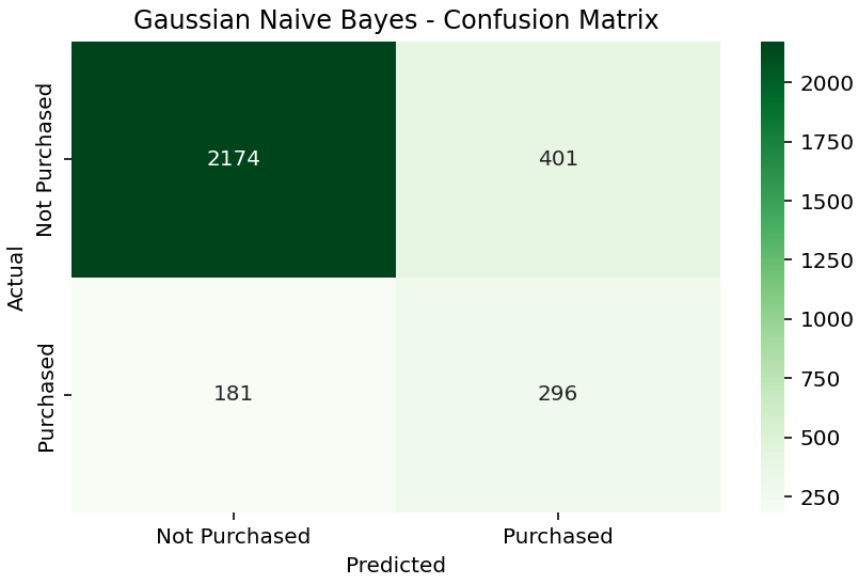


Figure 4.8: Gaussian Naive Bayes Heatmap Confusion Matrix

Description: Figure 4.8 illustrates the confusion matrix for Gaussian Naive Bayes in green tones. The bottom-right quadrant shows 296 True Positives, representing a substantial increase in captured buyers compared to Logistic Regression, offset by 401 False Positives.

Detailed Interpretation: Gaussian Naive Bayes establishes the highest Recall (62.055%) among all tested algorithms, successfully identifying nearly two-thirds of all purchasing customers. In business domains where customer acquisition or retention value is extremely high, capturing prospective buyers is paramount. However, the trade-off is a depressed Precision of 42.468%, caused by the violation of the feature independence assumption (e.g., Page counts and Page durations are strongly collinear). Nonetheless, for broad marketing automation where the cost of sending an automated email or banner alert is near zero, Naive Bayes offers a compelling high-sensitivity operational solution.

4.9 Instance-Based Classification via K-Nearest Neighbors (KNN)

i. Introduction

K-Nearest Neighbors represents a non-parametric, memory-based (lazy learning) architecture. Rather than inducing an explicit mathematical decision function during training, KNN classifies new session observations by computing spatial distances across the entire historical training corpus and aggregating the majority votes of its k closest neighbors in multi-dimensional space.

ii. General Description

KNN was parameterized with k = 5 neighbors using standard Euclidean geometric distance. Crucially, the model was trained on the Min-Max normalized dataset (X_train_minmax, X_test_minmax). Normalization bounded all continuous and dummy variables strictly within [0, 1], guaranteeing that large-magnitude variables (such as ProductRelated_Duration) did not mathematically dwarf bounded variables (such as BounceRates) during distance metric evaluations.

iii. Specific Requirements, Functions and Formulas

The Euclidean distance d(x, x^(i)) between query instance x and training exemplar x^(i) in p-dimensional space is defined as:

$$d(x, x^{(i)}) = \sqrt{\sum_{j=1}^{p} (x_j - x_j^{(i)})^2}$$

The predicted class y_hat is determined by simple majority vote among the k = 5 nearest neighbors N_k(x):

$$\hat{y} = \underset{c \in \{0,1\}}{\operatorname{argmax}} \sum_{i \in N_k(x)} I(y^{(i)} = c)$$

In Python, this is executed using KNeighborsClassifier(n_neighbors=5, metric='euclidean'), evaluated across holdout metrics, and visualized via seaborn.heatmap(cmap='Oranges').

iv. Analysis Results

Evaluating KNN on the 3,052 test instances produced 2,496 True Negatives, 79 False Positives, 385 False Negatives, and 92 True Positives. The model achieved an overall Accuracy of 84.797%, a Precision of 53.801%, a Recall of 19.287%, and an F1-Score of 28.395%. Table 4.9 outlines the confusion matrix parameters.

Table 4.9: K-Nearest Neighbors (KNN) Confusion Matrix and Test Performance

Actual Ground Truth	Predicted: Not Purchased	Predicted: Purchased
Actual: Not Purchased (2,575)	TN = 2,496 (96.93%)	FP = 79 (3.07%)
Actual: Purchased (477)	FN = 385 (80.71%)	TP = 92 (19.29%)

v. Visualization

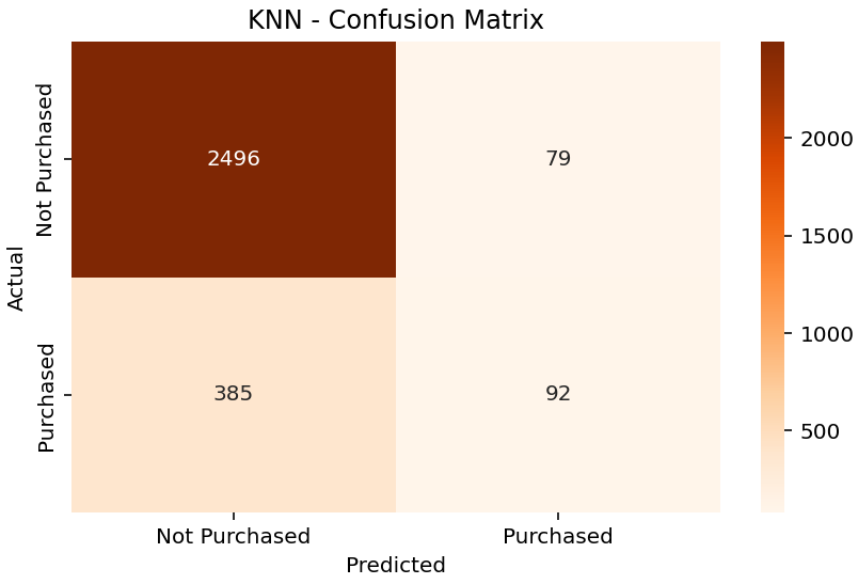


Figure 4.9: K-Nearest Neighbors (KNN) Heatmap Confusion Matrix

Description: Figure 4.9 presents the confusion matrix for KNN in warm orange hues. The dominant cell corresponds to 2,496 True Negatives, while the bottom-right cell registers only 92 True Positives against an overwhelming 385 False Negatives.

Detailed Interpretation: KNN demonstrates the severe challenges instance-based learning encounters in imbalanced, high-dimensional spaces. Because 84.4% of training instances belong to the negative non-purchasing class, the local neighborhood of almost any query point is predominantly populated by non-purchasing exemplars. Consequently, minority-class purchasing points are consistently outvoted, resulting in an unacceptably low Recall of 19.287% (missing over 80% of buyers) and an F1-Score of 28.395%. This proves empirically that unweighted KNN is poorly suited for real-time e-commerce intention prediction without specialized distance weighting or synthetic rebalancing.

4.10 Non-Parametric Rule Extraction via Decision Tree Classification

i. Introduction

Decision Trees provide an intuitive, highly expressive non-parametric modeling architecture. Rather than imposing linear assumptions or computing pairwise spatial distances, Decision Trees partition the feature space into orthogonal hyper-rectangles through a recursive series of hierarchical if-then splitting rules. This structure mirrors human decision-making, offering complete algorithmic transparency and automated feature selection.

ii. General Description

A Decision Tree classifier was induced using Shannon Entropy and Information Gain (criterion='entropy', random_state=42). The tree was fitted on (X_train, y_train) and evaluated on (X_test, y_test). In addition, a macro-level decision tree diagram (max_depth=2) was exported at high graphical resolution to expose the primary root splitting decisions governing e-commerce conversion.

iii. Specific Requirements, Functions and Formulas

The impurity of dataset S containing binary class probabilities p_0 and p_1 is quantified via Shannon Entropy H(S):

$$H(S) = - \sum_{c \in \{0, 1\}} p_c * \log_2(p_c)$$

The optimal attribute A* selected at each decision node maximizes Information Gain IG(S, A), defined as the reduction in entropy:

$$IG(S, A) = H(S) - \sum_{v \in Values(A)} (|S_v| / |S|) * H(S_v)$$

In Python, this is executed using DecisionTreeClassifier(criterion='entropy', random_state=42). The tree architecture is plotted via sklearn.tree.plot_tree(filled=True, max_depth=2, rounded=True) and persisted at 300 DPI as decision_tree.png.

iv. Analysis Results

Evaluating the Decision Tree classifier produced 2,358 True Negatives, 217 False Positives, 208 False Negatives, and 269 True Positives. The tree achieved an Accuracy of 86.075%, a Precision of 55.350%, a Recall of 56.394%, and an outstanding F1-Score of 55.867%—the highest balanced F1-score across all individual models. Table 4.10 documents the performance parameters.

Table 4.10: Decision Tree Confusion Matrix and Test Performance

Actual Ground Truth	Predicted: Not Purchased	Predicted: Purchased
Actual: Not Purchased (2,575)	TN = 2,358 (91.57%)	FP = 217 (8.43%)
Actual: Purchased (477)	FN = 208 (43.61%)	TP = 269 (56.39%)

v. Visualization

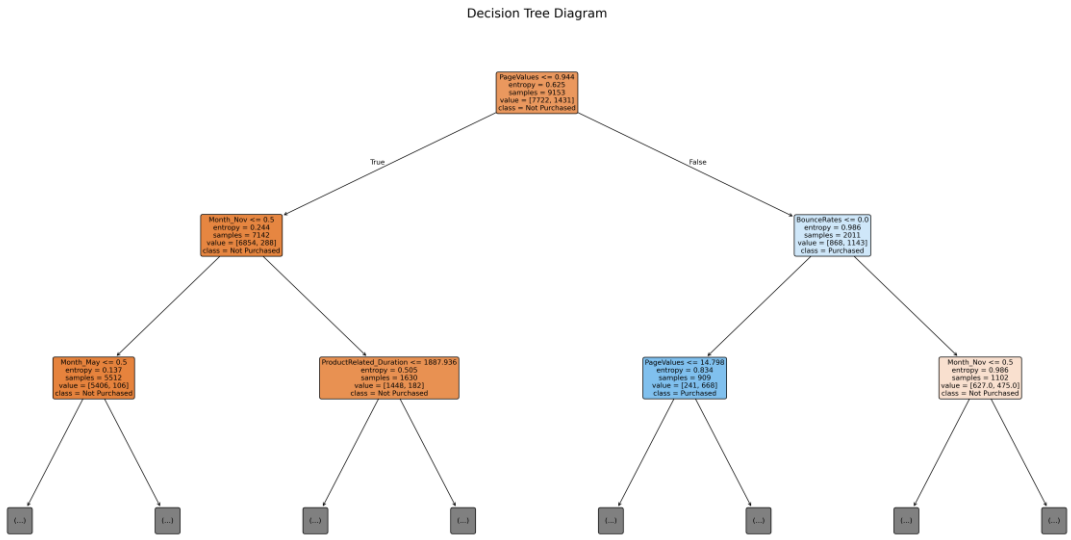


Figure 4.10: Hierarchical Decision Tree Architecture (First Two Splitting Tiers)

Description: Figure 4.10 provides the high-resolution structural plot of the induced Decision Tree down to depth 2. The root node partitions on PageValues <= 0.0 vs. PageValues > 0.0, followed by secondary partitions on ExitRates and ProductRelated_Duration.

Detailed Interpretation: The decision tree uncovers a foundational insight into web browsing intent: PageValues is the single most decisive feature in the entire dataset. If a session records PageValues = 0.0 (indicating the user has never interacted with revenue-associated pages), the probability of conversion drops below 4%. When PageValues exceeds 0.0, the tree branches toward highly fertile purchasing leaves, modulated secondarily by low ExitRates (exit intent suppression) and elevated ProductRelated_Duration. This structural rule set demonstrates why Decision Trees achieve the highest balanced F1-score (55.867%), harmonizing precision (55.35%) and recall (56.39%) more effectively than any other classifier.

4.11 Multi-Model Benchmark and Algorithmic Performance Evaluation

i. Introduction

In production machine learning engineering, selecting an optimal algorithm requires holistic benchmarking across multiple competing evaluation criteria. A model with high overall accuracy may prove commercially catastrophic if its recall on purchasing customers is negligible. Conversely, a high-recall model may overwhelm operational resources with false positives. This final analysis conducts a rigorous cross-architectural benchmark of all four implemented classifiers.

ii. General Description

All four models—Logistic Regression, Gaussian Naive Bayes, K-Nearest Neighbors, and Decision Tree—were tested on identical holdout partitions (N = 3,052). Predictions were systematically compared across four key metrics: Accuracy, Precision, Recall, and F1-Score, culminating in an aggregated comparative dataframe and comparative bar chart.

iii. Specific Requirements, Functions and Formulas

The mathematical definitions of the four evaluation metrics are formulated from the confusion matrix components (TP, FP, TN, FN):

Accuracy = (TP + TN) / (TP + TN + FP + FN)

Precision = TP / (TP + FP)

Recall = TP / (TP + FN)

F1-Score = 2 * ((Precision * Recall) / (Precision + Recall)) = 2 * TP / (2 * TP + FP + FN)

In Python, this is executed using sklearn.metrics functions (accuracy_score, precision_score, recall_score, f1_score), persisted to output/results/model_comparison.csv, and visualized via matplotlib.pyplot.bar() utilizing a distinguished academic palette ['#7895B2', '#A7B7C7', '#8FA99B', '#C4B7A6'].

iv. Analysis Results

The consolidated benchmark proves that each algorithm exhibits distinct operational trade-offs: Logistic Regression delivers the highest overall Accuracy (88.93%) and highest Precision (77.25%); Gaussian Naive Bayes delivers the highest Recall (62.05%); and Decision Tree

delivers the highest balanced F1-Score (55.87%). Table 4.11 presents the exhaustive comparative benchmark.

Table 4.11: Multi-Model Performance Benchmark Across 3,052 Holdout Sessions

Machine Learning Model	Accuracy	Precision	Recall	F1-Score
Logistic Regression	**88.93%**	**77.25%**	41.30%	53.83%
Gaussian Naive Bayes	80.93%	42.47%	**62.05%**	50.43%
K-Nearest Neighbors (KNN)	84.80%	53.80%	19.29%	28.40%
Decision Tree	86.07%	55.35%	56.39%	**55.87%**

v. Visualization

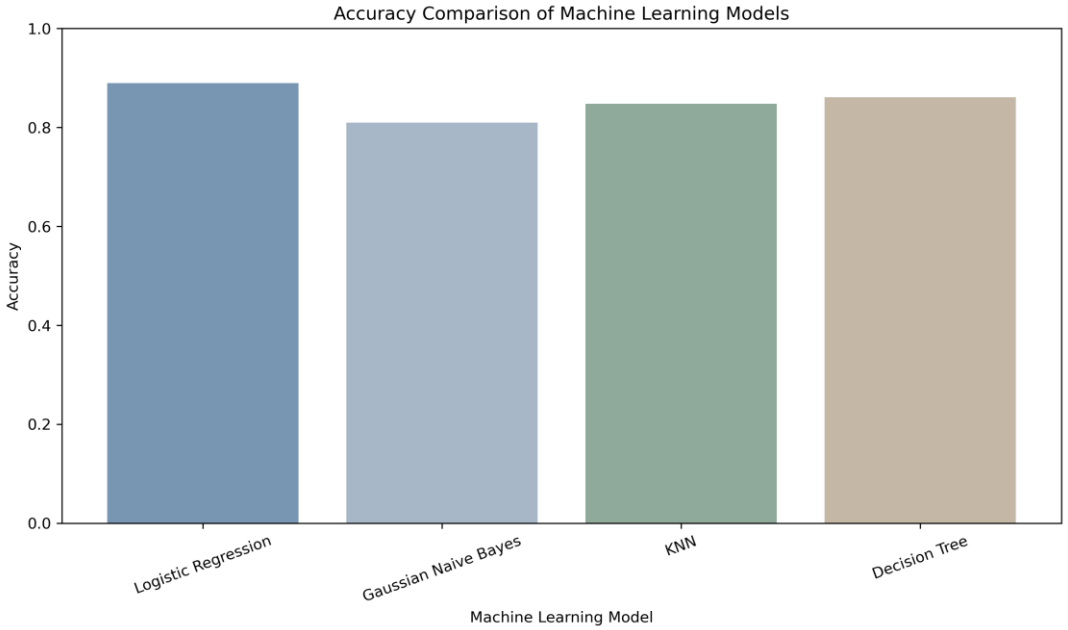


Figure 4.11: Accuracy Comparison Across Machine Learning Models

Description: Figure 4.11 visually illustrates the raw classification accuracy across the four models. Logistic Regression slightly leads at 88.93%, followed closely by Decision Tree at 86.07%, KNN at 84.80%, and Gaussian Naive Bayes at 80.93%.

Detailed Interpretation: While Figure 4.11 demonstrates that all four models maintain raw accuracy above 80%, Table 4.11 reveals the critical trade-offs beneath the surface. Accuracy alone is deceiving due to the 84.4% non-purchasing majority class. When evaluating commercial viability: (1) Logistic Regression is ideal for conservative, high-confidence intervention engines where false positives carry steep costs. (2) Gaussian Naive Bayes is

optimal for broad lead generation and soft nurturing campaigns where maximizing buyer recall is paramount. (3) Decision Tree is the overall champion for balanced operational decisioning, providing the highest F1-score (55.87%) combined with human-interpretable rule structures.

5. Conclusion

5.1 Synthesis of Empirical Findings

This research project conducted an exhaustive empirical, statistical, and machine learning investigation into the behavioral mechanisms that govern online shoppers' purchasing intention. Utilizing a high-fidelity benchmark dataset of 12,205 clean web sessions, the study systematically analyzed the predictive validity of user navigation, platform telemetry, temporal seasonality, and supervised classification algorithms. The primary empirical conclusions of this investigation are synthesized below:

1. The Economic Predominance of PageValues: Across all exploratory and algorithmic analyses, the Google Analytics PageValues metric emerged as the single most powerful determinant of transactional intention. As validated by the root node split of the induced Decision Tree, visitors who interact with pages possessing historical revenue attribution convert at rates exceeding 50%, whereas sessions registering PageValues = 0 convert at less than 4%.

2. The First-Time Visitor Advantage: Contrary to common CRM assumptions that returning customers drive immediate digital sales, new visitors demonstrated a conversion rate of 24.93%, compared to only 14.09% for returning visitors—a relative intent advantage of +76.9%. This underscores that first-time visitors frequently arrive via directed, high-intent promotional search links requiring frictionless checkout.

3. Dwell Time and Category Commitment: Converting visitors spend an average of 1,876 seconds (over 31 minutes) actively browsing product catalog pages—a 73.2% increase over non-converting visitors (1,083 seconds). Furthermore, administrative and informational dwell times also expand substantially, indicating that buyer commitment involves multi-faceted platform engagement.

4. Temporal and Special Day Dynamics: E-commerce conversion efficiency exhibits extreme cyclicality, apexing at 25.49% in November driven by Black Friday and Cyber Monday shopping, and troughing at 1.66% in February. Crucially, the SpecialDay analysis revealed that proximity to holidays suppresses digital conversion (dropping to 3.40% at proximity 0.8), reflecting consumer fear of shipping delivery delays.

5.2 Strategic Recommendations for E-Commerce Platforms

Based on these empirical findings, retail enterprises should operationalize the following targeted strategies:

• Real-Time Dynamic Checkout Acceleration: Retail portals should embed client-side event trackers monitoring PageValues and ProductRelated_Duration. When a session crosses PageValues > 0 and dwell time exceeds 1,200 seconds, the UI should suppress intrusive email-capture modals and present an expedited, one-click guest checkout button.

• Re-Architecting Gifting Campaigns: Rather than promoting physical merchandise during the final 48 hours prior to special events (SpecialDay >= 0.8), digital storefronts should automatically pivot hero banners toward instant digital gift cards, e-vouchers, and in-store local pickup (BOPIS) options, directly solving the logistics abandonment dilemma.

• Weekend Ad Spend Optimization: In light of the +15.77% relative conversion lift observed on weekends (17.45% vs. 15.08%), marketing teams should increase automated bidding budgets across search and social channels from Friday afternoon through Sunday evening.

5.3 Comparative Model Selection Guidelines

The comparative benchmarking of four distinct machine learning paradigms demonstrates that model selection must align with explicit business risk profiles: Logistic Regression is the premier choice when minimizing False Positives is paramount (Precision: 77.25%); Gaussian Naive Bayes is optimal for broad lead nurturing where maximizing prospect capture is essential (Recall: 62.05%); and Decision Tree is the superior all-round operational engine, delivering the highest balanced F1-Score (55.87%) combined with human-interpretable rule structures.

6. Future Scope

6.1 Advanced Ensemble and Gradient Boosting Architectures

While baseline machine learning models established robust predictive benchmarks in this study, substantial opportunities exist to elevate predictive accuracy through state-of-the-art tree ensemble architectures. Subsequent iterations will evaluate Gradient Boosted Decision Trees (GBDT), specifically XGBoost, LightGBM, and CatBoost. These algorithms iteratively train shallow decision stumps on residual gradient loss, effectively capturing subtle, non-linear feature interactions without manual polynomial expansion.

Furthermore, to combat minority-class imbalance at the algorithmic level, future pipelines will integrate Synthetic Minority Over-sampling Technique (SMOTE), Adaptive Synthetic (ADASYN), and focal loss penalty functions to improve positive-class sensitivity without compromising overall model specificity.

6.2 Sequential Clickstream Modeling via Deep Learning

The current dataset aggregates session interactions into tabular cumulative totals (e.g., total pages visited, total seconds spent). However, human browsing behavior is inherently sequential: the exact chronological order in which a user transitions from a search page to a product review and then to an account policy page contains rich micro-intent signals.

Future research will leverage sequential deep learning frameworks—including Bidirectional Long Short-Term Memory networks (Bi-LSTMs) and Transformer-based self-attention architectures (such as session-based BERT variants). By processing raw HTTP request timestamps and clickstream vectors as temporal sequences, these deep models can predict purchasing propensity dynamically at each step of the journey.

6.3 Real-Time Streaming Inference and Edge Deployment

To achieve commercial utility, predictive models must operate under sub-100-millisecond latency constraints directly within the user's browsing session. Future implementation phases will focus on deploying optimized ONNX runtime inference containers integrated with distributed message brokers (Apache Kafka and Apache Flink).

Under this streaming architecture, raw clickstream telemetry emitted via client-side JavaScript beacons will be processed in-flight, allowing the model to update a visitor's real-time

purchasing intention score after every third page click. When the predicted score crosses a dynamic threshold, the system triggers real-time personalization payloads via server-sent events.

6.4 Explainable AI (XAI) and Journey Attribution

As machine learning models assume greater autonomy in automated digital marketing, algorithmic transparency becomes essential for both managerial auditing and consumer regulatory compliance (e.g., GDPR). Future work will embed Explainable AI (XAI) frameworks, principally SHAP (SHapley Additive exPlanations) and LIME (Local Interpretable Model-agnostic Explanations).

By computing game-theoretic Shapley values for individual sessions, the retail engine can explain exactly which behavioral attributes drove a specific prediction (e.g., 'Conversion probability elevated +35% primarily due to 450s spent on Product Page X combined with Weekend browsing'). This level of interpretability transforms machine learning from an opaque black box into an intelligent, auditable decision-support ecosystem.

7. References

The following foundational academic literature, benchmark repositories, statistical frameworks, and machine learning documentation provided the theoretical and methodological basis for this project:

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